

Filesystem Maintenance

René Serral-Gracià¹

¹Universitat Politècnica de Catalunya (UPC)

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Lectures

- 1 System administration introduction
- 2 Operating System installation
- 3 User management
- 4 Application management
- 5 System monitoring
- 6 **Filesystem Maintenance**
- 7 Network services
- 8 Security and Protection
- 9 Introduction to Public Cloud

Outline

- 1 Introduction
 - Goals
- 2 Filesystems
- 3 Disk Handling
- 4 Logical Volume Manager (LVM)
- 5 Backups

Goals

Knowledge

- Filesystems
- Backup tools
- Backup media

Abilities

- Filesystem resizing
- Filesystem verification
- Perform and restore backups

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Filesystems (I)

- FAT (FAT16) → DOS
 - Small disks (< 4GB)
 - File names 8+3
- FAT32 (VFAT) → Win95
 - Larger disks
 - Long filenames
 - Partial definition of soft-links
 - No owner or file access privileges
- exFAT
 - FAT32 Extension
 - Theoretical maximum capacity of 64ZiB (512TiB real)
- NTFS → WinNT, XP, Vista, Windows 7
 - Integrates ownership and privileges (create, modify, access...)
 - Maps to Windows NT security model

Filesystems (II)

- ext2
 - UNIX Filesystem
 - Soft/hard links
 - Access privileges
 - Long filenames
- ext3
 - Adds journaling (eases error recovery)
- reiserfs
 - Files and directories organized similarly to a database
 - Features journaling
 - Very efficient in small files
 - No internal block fragmentation

Filesystems (III)

- xfs
 - journaling
 - Dynamic i-node management
 - ACLs
 - Very large disk sizes
 - Filesystem activity log
- jfs
 - journaling
 - Dynamic i-node management
 - ACLs and MAC (Mandatory Access Control)
 - Very large disk sizes

Filesystems (and IV)

- ext4
 - 64 bits addressing, improvements in journaling
 - Delayed allocation
 - Extents
 - 1 exbibyte (EiB) maximum size
- btrfs
 - Extents
 - Online resizing
 - Online balancing
 - Online filesystem check

Journal filesystems

- Journal: disk operation registry
 - Eases the recovery of the FS in case of crash or error
 - Slightly decrease in disk operations performance
- Journal outside the buffer cache
 - Journal can be stored in another disk or partition
- Ext3/4, reiserfs, JFS, XFS, NTFS, BTRFS have journal

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- 1 Introduction
- 2 Filesystems
- 3 Disk Handling**
 - Disk Verification
 - Extending disk space
 - Disk quota management
 - Other Management tasks
- 4 Logical Volume Manager (LVM)
- 5 Backups

Disk verification

- Process to check a drive for errors
- Detecting errors early may avoid data loss

Reasons

- Hardware errors
- Power shortage
- Operating system bugs
- Administration errors
 - Incorrect machine shutdown

Never verify a mounted filesystem

- High probability of disk corruption
- Verification access skips the buffer cache and acts directly on the device

Types of Disk verification

Logical verification

- Filesystem metadata
- Directory structure
- Lost data recovery
 - Directory `lost+found`

Physical recovery

- Disk blocks with Input/output errors
- Command: `badblocks`

What to do when a partition gets full

- We can buy another disk and integrate it into the system
- We could try to increase the partition size
 - **Danger of data loss**

Install and configure the new disk

- Create filesystem and decide the mount points
- Transfer the required data to the new partition (onto a temporary mount point)
- Re-Mount the partition to the final destination
 - Update `/etc/fstab`
- Maybe you have to reorganize existing directories
 - `/home` → `/homeA` + `/homeB`
 - `/home` → `/home/students` + `/home/professors`

Exercise

Plan and issue a filesystem resizing for the partitions

- /home
- /var

Disk quota (I)

Quota

Ability to limit the amount of data a user (or user group) is able to use in a filesystem (partition)

Requires

- Support from the filesystem
- Support from the kernel

Quota management (II)

Partition preparation

- Mounted using options 'usrquota' and/or 'grpquota'
- It can be done from /etc/fstab

```
/dev/sda3 /home ext4 defaults,usrquota,grpquota 1 1
```

- quotacheck command to create the quota files

```
quotacheck -v -a -g -u -m  
verbose all group user no-remount
```

- Creates
 - /aquota.user
 - /aquota.group

Quota management (III)

- Quota enabling

```
quotaon -v      -a -g -u
           verbose all group user
```

- Activates quota mechanisms, usually from `/etc/init.d/`

- Quota disabling

- `/sbin/quotaoff`

- Quota editing (`edquota`)

```
Disk quotas for user rserral (uid 1000):
Filesystem    blocks      soft      hard      inodes    soft      hard
/dev/sda3      10246    1024716    2024716         0         0         0
```

- Data blocs and i-nodes quota
- It is not possible to edit the used blocks/inodes, but the limits of the quota can be changed

Quota management (and IV)

- Visualize quotas: `quota -v`

```
Disk quotas for user rserral (uid 1000):
```

Filesystem	blocks	quota	limit	grace	files	quota	limit	grace
/dev/sda3	1324716*	1024716	2024716	6days	145	0	0	-

* We are above the quotas, within the "hard" limit!!

- "Grace period"
 - Grace time where the user can reach the hard limit, it only raises warnings
 - If the grace period expires, then the system does not allow to go above the soft limit

Other maintenance tasks

Monitoring

- Free space (`df`)
 - Most systems reserve a (5%) of the space to be exclusively used by root
- Occupied space (`du`)

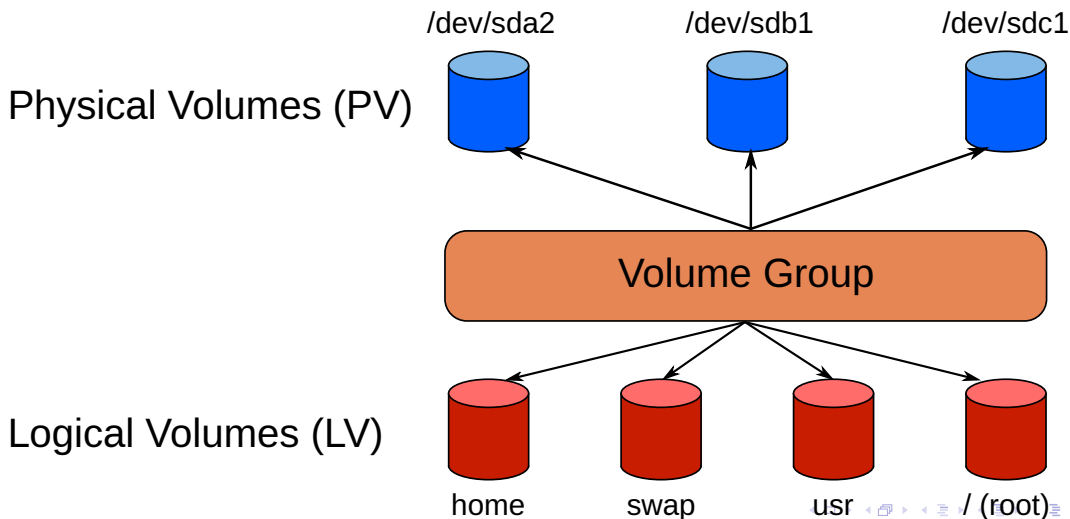
Synchronization

- Write to disk the modified buffers
 - `sync`
 - Update daemon

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Logical Volume Manager (LVM) (I)



Logical Volume Manager (and II)

- High level abstraction of the disk space
- Aggregates multiple physical partitions/disks
 - Allows to add more partitions to the volume
- It allows logical partitions within the volume
 - They can be assigned logical names
 - Customized distribution among the physical volumes
 - Resizing
 - Move
- Example: `/etc/fstab`

```
/boot    /dev/sda1    ...  
swap     /dev/vg00/swap ...  
/         /dev/vg00/root ...  
/home    /dev/vg00/home ...  
/usr     /dev/vg00/usr ...
```

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 - Full Backup
 - Incremental Backup
 - Reverse Incremental Backup
 - Differential Backup

Backups

Preliminary considerations

- Data to copy
 - User data (home, mail)
 - Program data (DBs, GIT, web, ...)
 - System configuration
 - Binary?
- Backup types
 - Full Backup (all)
 - Incremental Backup (only changes)
 - Reverse Incremental Backup (only changes)
 - Differential Backup
- Backup frequency
 - Data confidence
 - Data importance

Backups

Physical support

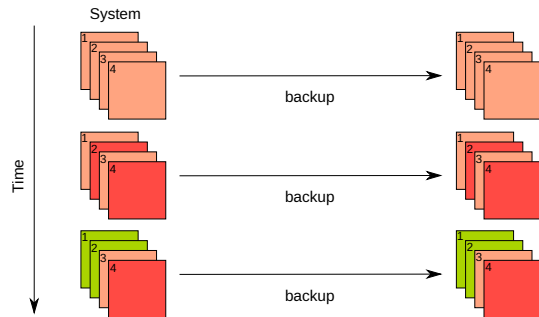
- Floppy, disc, CD, tape, network. . .
- To consider:
 - Cost size ratio
 - Reliability
 - Availability
 - Usability
 - Speed

Copy location

- Accident protection
- Fireproof boxes
- Keep some backups outside the company premises
- Stealing protection

Full Backup

- Always copy all the data
 - Fast to restore
 - Large size
- It is the base for all backup types
- Can be compressed
- Can be used on any storage medium



Incremental Backup

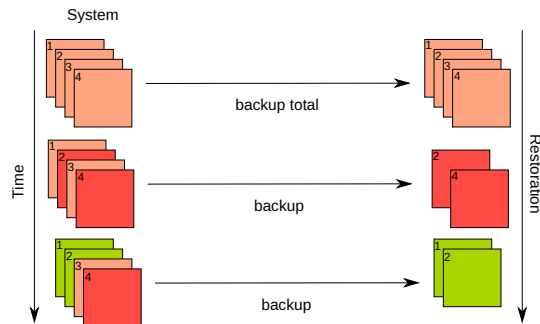
- It only backs up the changed files
- Can be compressed
- Can be used on any storage medium

Advantages

- Smaller size

Inconveniences

- It is slower to restore
- Tedious with long chains



Reverse Incremental Backup

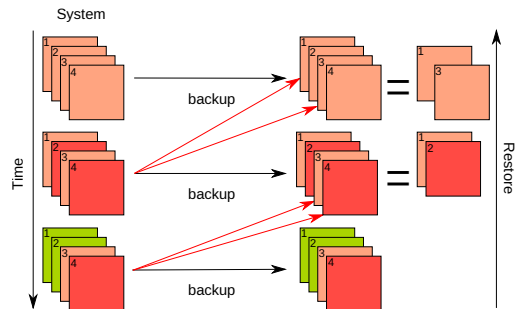
- Everything is copied (linked) from the previous backup
- Except for the modified files
 - Which are taken from the original data

Advantages

- Fast to restore
- Little space

Inconveniences

- Only random access storage



Differential Backup

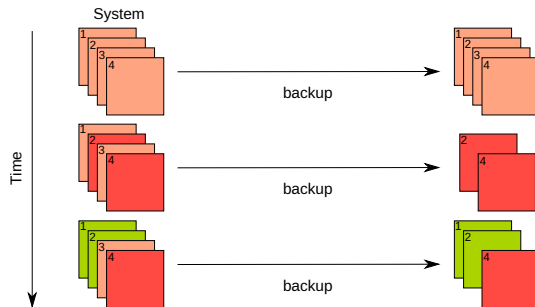
- All changes since the last full backup are copied

Advantages

- Allows to remove Incremental
- Allows a shorter backup chain

Inconveniences

- Large space needs
- Admin should erase incremental



Backups – Tapes

	LTO-1	LTO-2	LTO-3	LTO-4	LTO-5	LTO-6	LTO-7	LTO-8	LTO-9	LTO-10	
Release date	2000	2003	2005	2007	2010	2012	2015	2017	TBA	TBA	
Native/raw data capacity	100 GB	200 GB	400 GB	800 GB	1.5 TB	2.5 TB	6.0 TB	12 TB	26 TB	48 TB	
Max uncompressed speed (MB/s)	20	40	80	120	140	160	300	360	708	1100	
Time to write a full tape at max uncompressed speed(hh:mm)	1:25	1:25	1:25	1:50	3:10	5:30	5:50	8:45	10:40	12:40	
Compression capable?	Yes, "2:1"					Yes, "2.5:1"			Planned, "2.5:1"		
WORM capable?	No		Yes						Planned		
Encryption capable?	No			Yes						Planned	
Max. number of partitions	1 (no partitioning)				2		4			Planned	

¹Source: https://en.wikipedia.org/wiki/Linear_Tape-Open

Exercise

Define a backup policy:

- Data to backup
- Backup type(s)
- Frequency
- Storage Media
- Compression
- Expiration policy

For a server with:

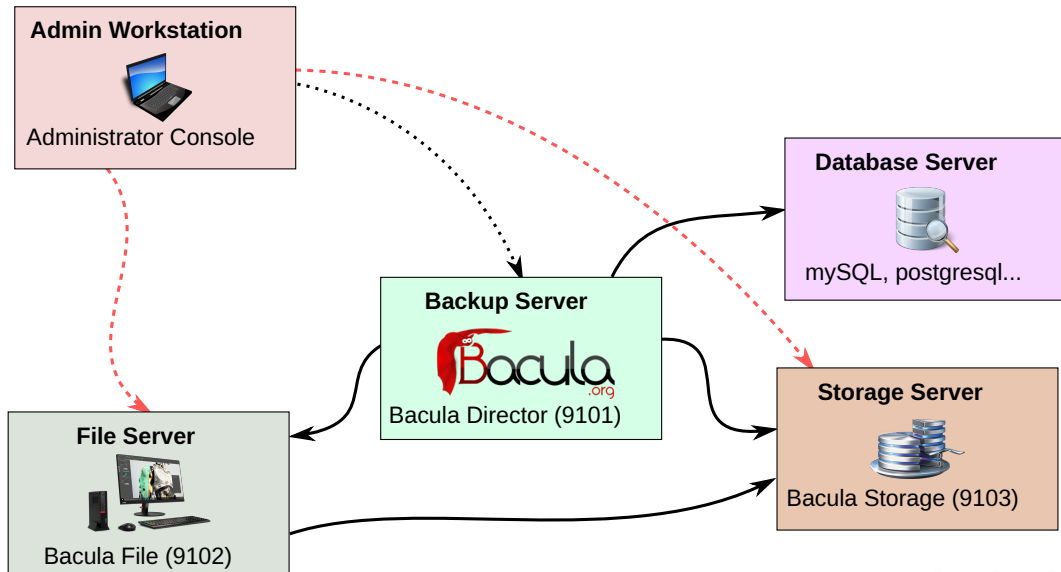
- **Users:** 50 Gb. disk per user (80 users)
- **Mail:** 1 Gb per user
- **Web pages**
 - 2 Gb Main site
 - 100Gb Database
- **Code repository**
 - 100 GB distributed among 10 projects
 - Only 5 active projects

Other considerations

- Dedicate specific machines for backup
 - It takes many resources
 - easier to administer
- Consider how much data you want to potentially loss
- Consider how much storage you will need
- Be aware of legislation

Tools:

bacula, Borg



Personal Homework

- Task automation
 - Programming language: `bash`, `perl`
 - Information search: `find`, `grep`...